

Analysis

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Included with the class material are different methods of analysing sound data (*amplitude, frequency, spectral characteristics, onsets etc.*)

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Sound streams can be *analysed, measured* and *treated* in ways that we use information from these to drive *synthesis* or *compositional choices*.

Possible considerations regarding sound analysis:

- * Continuous / Discrete
- * Client / Server (SC).
- * Realtime / Offline
- * Accuracy / Features
- * Perception / Hard Data

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Table 15.1
Overview of Machine-Listening Resources for SuperCollider

Category	UGen	Description	Availability (core unless otherwise stated)
Amplitude	Amplitude RunningSum Loudness	RMS amplitude perceptual loudness	
Pitch	ZeroCrossing Pitch Tartini Qitch	simplistic pitch detection; also as a timbral feature autocorrelation in time domain adaptation from Phil McLeod's Tartini program constant Q transform pitch detector after Judith Brown and Miller Puckette	Nick Collins (NC) NC
Timbre	SpecCentroid SpecFlatness SpecPcile MFCC	spectral features correlating, respectively, with perceived brightness, noisiness, and roll-off point Mel Frequency Cepstral Coefficients	
Onset detection	PV_HainsworthFoote PV_JensenAndersen Onsets	Dan Stowell's onset detector encapsulates his research work including many different functions	
Event Extraction	AnalyzeEvents2		NC
Beat tracking	BeatTrack BeatTrack2 DrumTrack	Autocorrelation beat tracker Crosscorrelation beat tracker Drum pattern template detector	NC NC
Key extraction	KeyTrack		
Psychoacoustics	Dissonance classes	Language-side classes for sensory dissonance analysis	Juan Sebastián Lach

Cross-Synthesis

Cross Synthesis

“Cross-synthesis is the technique of impressing the spectral envelope of one sound on the flattened spectrum of another. A typical example is to impress speech on various natural sounds, such as ``talking wind.” Let's call the first signal the ``modulating” signal, and the other the ``carrier” signal. Then the modulator may be a voice, and the carrier may be any spectrally rich sound such as wind, rain, creaking noises, flute, or other musical instrument sound.”

(<http://www.dsprelated.com>)

Exercises

Analysis - Exercises

1. Implement a synthesis model that uses frequency and loudness analysis from another sound for controlling its own. The information amount should be adjustable as the process unfolds.
2. Implement an onset detector that generates rhythmic synthetic sound from analysed events.
3. Implement chords of synthetic sound where each of the partials are obtained by analysing an input sound.
4. Implement a small vocoder that analyses sound in different frequency areas and applies these to a rich oscillator signal.

Cross Synthesis - Exercises

1. Choose two sounds for cross synthesis. One should be rich and constant, the other to contain dynamics and audible changes. Cross synthesize the two where the dynamic sound will modulate the more rich and constant one.
2. Repeat the process in exercise 1. but reverse the input sounds. Additionally transform the input sounds before executing the cross synthesis (for example by time stretching or extending) so that the process will be more effective.
3. Implement a transformation from sound A to sound B that uses cross synthesis as a way of bridging material. The transformation could occur in steps or continuously.

